

Software Requirements Specification

Club Artizen Social Media Analytics Dashboard

Project: Club Artizen Dashboard | **Team:** 35

Team Members: Kaushik, Devshree, Kushagra, Sian, Srinivas

1. Brief Problem Statement

Club Artizen, a platform connecting artisans, creators, and community members, currently manages its presence across multiple social media channels (LinkedIn, Facebook/Instagram, WhatsApp, YouTube, and others) without a unified way to monitor how that content is actually performing. The marketing team and co-founders must log into each platform individually, manually collect data, and piece together a picture of their reach and engagement.

This is not just time-consuming, it also means that many patterns go unnoticed, opportunities to optimize content timing or format are missed, and there is no early warning when something is underperforming. The goal of this project is to build a centralized Social Media Analytics Dashboard that aggregates data from all of Club Artizen's channels into one dashboard, surfaces meaningful insights, and alerts the team when something important changes so they can focus on creating great content instead of dealing with numbers manually.

2. System Requirements

The following outlines the technical environment and dependencies required to build and run the Club Artizen Analytics Dashboard.

2.1 Platform & Infrastructure

- Web application accessible via modern desktop and mobile browsers.
- Backend server to handle API calls, data storage, and scheduled data.
- Cloud-hosted database for storing historical metrics and user configurations .

2.2 External APIs & Integrations

- META Graph API for Facebook and Instagram post metrics, page insights, paid/organic views, and audience demographics.
- LinkedIn Marketing API (Community Management API) for post performance, follower demographics, and page traffic metrics from both Company and Co-Founder pages.
- WhatsApp Business API (WABA) for messaging performance, conversation analytics, and template metrics. Note: SmartPing API is limited to message sending only and cannot retrieve analytics.
- YouTube Data API for video performance metrics (views, watch time, engagement).
- Email Campaign platform API.
- Google analytics API.
- Blog metrics.

2.3 Technology Stack

- Frontend: React.js or equivalent, with a drag-and-drop widget library for the customisable dashboard.
- Backend: Python.
- Database: Relational DB (MongoDB) for structured metrics; optional time-series layer for trend data.
- Authentication: JWT-based session for internal users.
- Hosting: Cloud platform (AWS / GCP / Azure TBD).

3. User Profile

These are the primary day-to-day users of the dashboard. They have solid digital marketing knowledge and understand concepts like impressions, CTR, engagement rate, and audience demographics. They need detailed, channel-level breakdowns and will use the analytics and predictive sections regularly. They should be able to build custom widgets without any coding knowledge. They will not have access to restricted Co-Founder page data unless explicitly granted.

4. Feature Requirements (Use Cases)

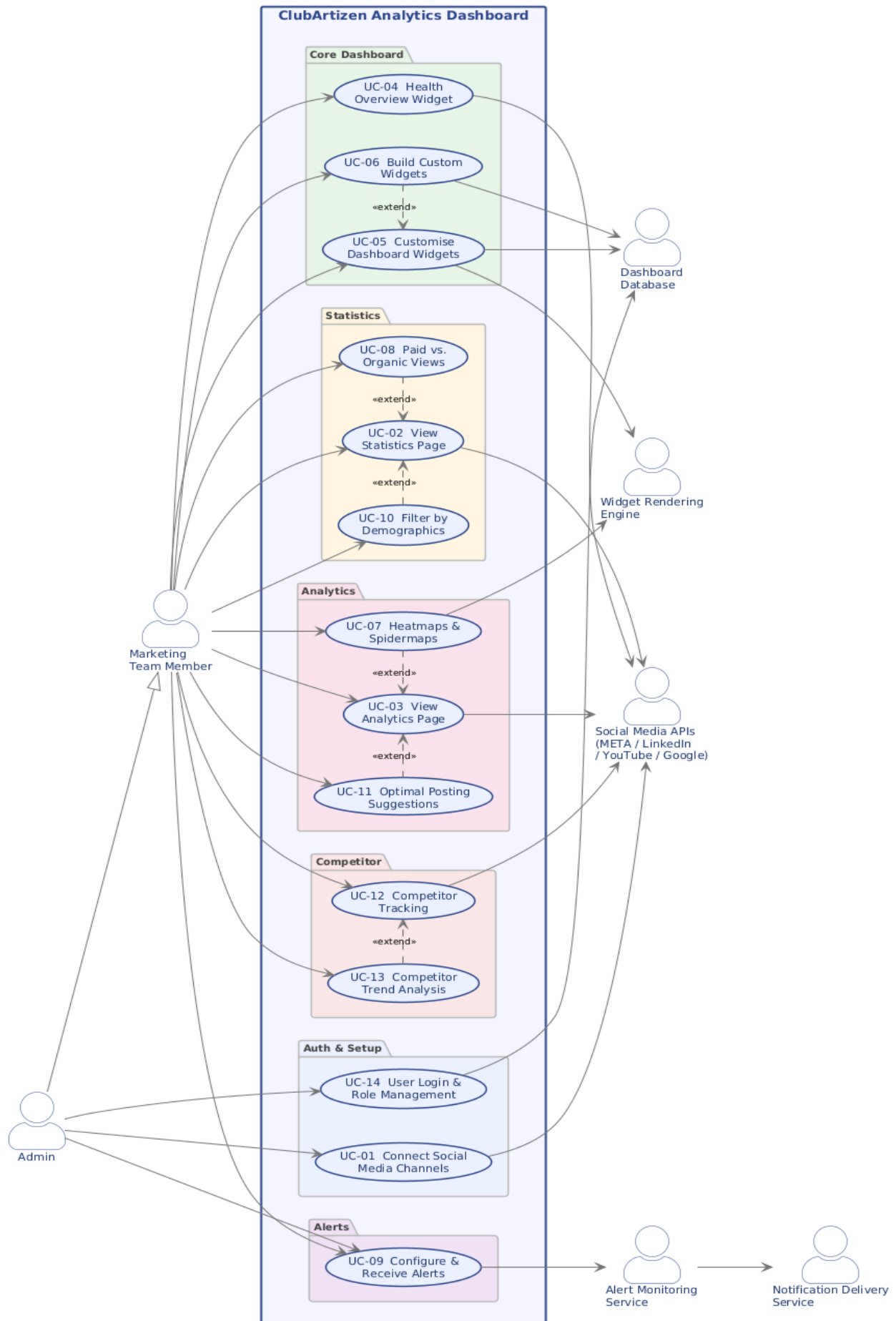
The table below lists the core use cases that define the features of the system. Each use case will be assigned to Release 1 (R1) or Release 2 (R2) during release planning.

No.	Use Case Name	Description	MoSCoW	Release
1	Connect Social Media Channels	The user (admin/founder) connects Club Artizen's social media accounts by providing credentials. The system authenticates and stores tokens securely, and confirms successful connection. This will be used to get all other data features.	Must Have	R1
2	View Statistics Page	The user views a Statistics page showing performance data per social media channel with follower growth graph, engagement metrics (likes, comments, shares), impressions/views, and click-through rate. Data must be comprehensive and support filtering by date range.	Must Have	R1
3	View Analytics Page	The user views an Analytics page that presents derived insights from the raw data including correlations (like posting time vs. performance, location vs. performance) and categorisation of top post types (Reel, Post, etc) and more.	Must Have	R1
4	Social Media Health Overview Widget	A central widget on the main dashboard shows a color-coded performance indicator for each social media channel. It gives users an overview status of how each channel is performing relative to defined thresholds.	Must Have	R1

5	Customise Dashboard with Widgets	The user can browse a library of pre-built data widgets and drag them onto their dashboard to create a personalised view. Widget placement and configuration is saved per user.	Could Have	R1
6	Build Custom Widgets	The user can create a custom widget by selecting up to two parameters (e.g. 'Engagement Rate' and 'Post Type') to define a unique chart or table. This widget creator must be usable by non-technical staff.	Should Have	R2
7	View Heatmaps and Spidermaps	The user can access heatmap and spidermap visualisations for applicable metrics for e.g., posts and click through rates , product types and performance.	Must Have	R2
8	View Paid vs. Organic Performance	The user can distinguish between paid and organic reach/views within the Statistics page, allowing the marketing team to evaluate the ROI of paid campaigns versus organic content performance.	Must Have	R1
9	Configure and Receive Alerts	The system monitors key thresholds and notifies users when engagement drops. Alert delivery method (dashboard + email / message) is yet to be confirmed.	Must Have	R2

10	Filter Data by different criterias	The user can filter statistics and analytics views by audience demographic (e.g. geography, age, company size) to understand which audience segments are giving more engagement.	Must Have	R2
11	Optimal Posting Suggestions	The system analyses historical performance data to suggest optimal posting times and recommended post lengths for each channel. Suggestions are surfaced in the Analytics or a dedicated Recommendations section.	Could Have	R2
12	Competitor Tracking	The user can add competitors to a tracked list. The system retrieves publicly available information about competitor channels. Note: Due to social media privacy policies (especially META), detailed metrics like clicks and views on competitor posts cannot be retrieved; only publicly visible signals are available.	Should Have	R2
13	Competitor Trend Analysis	Rather than real-time competitor metrics (which are largely inaccessible via APIs), the system surfaces trends in competitor content such as the types of posts they are publishing most frequently to help inform ClubArtizen's own content strategy.	Could Have	R2
14	User Login and Role Management	Authenticated users can log into the dashboard. Admins can assign roles (Co-Founder, Marketing Team) to control which data each user can view. Role permissions enforce the RBAC policy defined in Section 3.	Must Have	R1

ClubArtizen Social Media Analytics Dashboard — Use Case Diagram



6. Detailed Use Case Descriptions

The following section provides detailed descriptions for high-priority use cases.

UC-01: Manage Social Media Channels

Use Case Number	UC-01
Use Case Name	Manage Social Media Channels
Overview	Allows an authorised user to link one or more of ClubArtizen's social media accounts to the dashboard by providing OAuth credentials or API keys. Once connected, the system can begin fetching data from that channel.
Actors	Admin
Pre Condition	The user is logged in. The relevant developer account access and API credentials have been obtained from META, LinkedIn, Google, and/or WhatsApp Business.
Main Flow	1. User navigates to the 'Integrations' or 'Settings' page. 2. User grants permission / enters credentials. 3. System securely stores the access token or API key (encrypted). 4. System confirms successful connection and begins initial data fetch. 5. Channel appears as 'Connected' on the dashboard.
Alternate Flows	A1 Authentication fails: System displays an error message and prompts the user to retry or contact the platform's developer console. A2 Token expires: System detects an expired token, prompts the user to re-authenticate, and updates the stored credential.
Post Condition	The selected social media channel is connected, tokens are stored securely, and the channel's data is available for display across all dashboard views.

UC-02: View Statistics Page

Field	Detail
Use Case Number	UC-02
Use Case Name	View Statistics Page
Overview	Allows a user to view a dedicated Statistics page presenting raw performance data for each connected social media channel, including follower growth, engagement metrics, impressions, and click-through rates, with support for date range filtering.
Actors	Admin, Co-Founder, Marketing Team
Pre Condition	The user is logged in. At least one social media channel is connected (UC-01). Data has been successfully fetched from the connected channel(s).
Main Flow	1. User navigates to the 'Statistics' page from the main navigation. 2. System displays performance data for all connected channels. 3. User selects a specific channel to view its individual statistics. 4. System displays follower growth graph, engagement metrics (likes, comments, shares), impressions/views, and click-through rate. 5. User applies a date range filter to narrow the data displayed. 6. System refreshes the visualisations to reflect the selected date range.
Alternate Flows	A1 – No data available: If a channel has been connected but no data has yet been fetched, the system displays a loading state or a message indicating data is being retrieved.

Post Condition

A2 – Partial data: If only some metrics are available for a channel (e.g. WhatsApp Business has limited public metrics), the system displays available data and flags unavailable metrics clearly.

The user has viewed up-to-date raw performance statistics for one or more connected social media channels, filtered to their chosen date range.

UC-03: View Analytics Page

Field	Detail
Use Case Number	UC-03
Use Case Name	View Analytics Page
Overview	Allows a user to view an Analytics page that surfaces derived insights from raw statistics data, including correlations between posting time and performance, location-based performance, and categorisation of top-performing post types.
Actors	Admin, Co-Founder, Marketing Team
Pre Condition	The user is logged in. At least one social media channel is connected and has sufficient historical data to generate meaningful insights.

Main Flow	1. User navigates to the 'Analytics' page from the main navigation. 2. System processes raw data and derives insights, including posting time vs. engagement correlations, top-performing post types (Reels, Posts, Stories, etc.), and location-based audience performance. 3. System displays insights in visual format (charts, tables, ranked lists). 4. User can select a specific channel or view cross-channel insights. 5. User can apply filters (e.g. date range, post type) to refine the insights displayed.
Alternate Flows	A1 – Insufficient data: If there is not enough historical data to generate a reliable correlation or insight, the system displays a message indicating more data is needed and suggests a timeframe to return.
Post Condition	The user has reviewed derived analytics insights and can use these to inform content strategy and marketing decisions.

UC-04: Social Media Health Overview Widget

Field	Detail
Use Case Number	UC-04
Use Case Name	Social Media Health Overview Widget
Overview	A persistent widget displayed on the main dashboard provides a colour-coded performance indicator for each connected social media channel, giving users an at-a-glance overview of channel health relative to predefined performance thresholds.
Actors	Admin, Co-Founder, Marketing Team

Pre Condition	The user is logged in. At least one social media channel is connected and has data available. Performance thresholds have been configured (either by default or by an admin).
Main Flow	1. User lands on or navigates to the main dashboard. 2. System retrieves the latest performance data for each connected channel. 3. System compares each channel's metrics against defined thresholds. 4. Widget displays each channel with a colour-coded status indicator (e.g. green = performing well, amber = needs attention, red = underperforming). 5. User can hover over or click a channel indicator to see a brief summary of which metrics are driving the status.
Alternate Flows	A1 – Channel disconnected: If a previously connected channel loses its connection, its indicator displays a grey or warning state and prompts the user to reconnect. A2 – No thresholds set: If thresholds have not been configured, the system uses default benchmark values and notifies the admin that custom thresholds can be configured.
Post Condition	The user has a clear, colour-coded overview of each channel's current health status without needing to navigate to the full Statistics or Analytics pages.

UC-05: Customise Dashboard with Widgets

Use Case Number	UC-05
Use Case Name	Customise Dashboard with Widgets
Overview	The user browses a library of pre-built data widgets and drags them onto the main dashboard canvas to build a personalised view. The dashboard layout is saved and persists across sessions.

Actors	Co-Founder, Marketing Team Member
Pre Condition	The user is logged in. At least one social media channel has been successfully connected (UC-01).
Main Flow	1. User navigates to the main dashboard. 2. User opens the Widget Library panel. 3. User browses available widgets (filtered by channel, metric type, or visualisation type). 4. User drags a widget from the library onto the dashboard canvas. 5. System renders the widget with live data. 6. User can resize or reposition widgets by dragging. 7. User saves the layout; system persists the configuration.
Alternate Flows	A1 No data available for a widget: System displays a placeholder with a message indicating that data is not yet available for the selected channel. A2 User removes a widget: User drags it off the canvas or clicks 'Remove'; layout updates accordingly.
Post Condition	The user's personalised dashboard is saved and will be restored on their next login. All active widgets display up-to-date data.

UC-06: Build Custom Widgets

Field	Detail
Use Case Number	UC-06
Use Case Name	Build Custom Widgets
Overview	Allows a user to create a bespoke dashboard widget by selecting up to two parameters (e.g. Engagement Rate and Post Type) to define a unique chart or data table. The interface is designed to be accessible to non-technical staff.

Actors	Admin, Marketing Team
Pre Condition	The user is logged in with a role that has permission to create custom widgets. Sufficient data exists for the chosen parameters.
Main Flow	1. User navigates to the widget library or dashboard customisation panel and selects 'Create Custom Widget'. 2. System displays a guided widget builder interface. 3. User selects a primary parameter from a dropdown list (e.g. Engagement Rate). 4. Optionally, user selects a secondary parameter (e.g. Post Type) to create a comparative or correlated view. 5. User selects a preferred visualisation type (e.g. bar chart, line graph, table). 6. System generates a live preview of the widget using available data. 7. User names the widget and saves it. 8. The custom widget becomes available in the user's widget library for use on the dashboard.
Alternate Flows	A1 – Incompatible parameters: If the two selected parameters cannot be meaningfully combined, the system disables incompatible visualisation types and displays a tooltip explaining why. A2 – Insufficient data: If there is not enough data to populate the widget preview, the system notifies the user and suggests they return once more data has been collected.
Post Condition	A custom widget has been created and saved to the user's widget library, ready to be placed on the dashboard.

UC-07: View Heatmaps and Spidermaps

Field	Detail
Use Case Number	UC-07
Use Case Name	View Heatmaps and Spidermaps
Overview	Allows a user to access heatmap and spidermap visualisations for applicable metrics such as posting activity vs. click-through rates, and product type performance across multiple dimensions.
Actors	Admin, Co-Founder, Marketing Team
Pre Condition	The user is logged in. Sufficient data has been collected across the relevant metrics to generate meaningful heatmap or spidermap outputs.
Main Flow	1. User navigates to the Analytics page or a dedicated Visualisations section. 2. User selects 'Heatmap' or 'Spidermap' from the available visualisation options. 3. User selects the metric(s) to visualise (e.g. post frequency vs. click-through rate for a heatmap; multi-dimension performance across channels for a spidermap). 4. System renders the selected visualisation using available data. 5. User can hover over data points to view specific values. 6. User can apply date range or channel filters to refine the visualisation.
Alternate Flows	A1 – Insufficient data density: If there is not enough data to produce a meaningful heatmap (e.g. too few posts in the selected period), the system notifies the user and recommends a broader date range.
Post Condition	The user has viewed a heatmap or spidermap visualisation and can use it to identify patterns in posting behaviour, engagement, or product performance.

UC-08: View Paid vs. Organic Performance

Field	Detail
Use Case Number	UC-08
Use Case Name	View Paid vs. Organic Performance
Overview	Allows a user to distinguish between paid and organic reach and views within the Statistics page, enabling the marketing team to evaluate the ROI of paid campaigns relative to organic content.
Actors	Admin, Marketing Team
Pre Condition	The user is logged in. At least one connected channel has both paid and organic content data available via the relevant API.
Main Flow	1. User navigates to the Statistics page. 2. User selects a connected channel that supports paid content data (e.g. Meta, LinkedIn). 3. System displays a toggle or segmented view allowing the user to switch between 'Paid', 'Organic', and 'Combined' views. 4. System renders metrics (reach, impressions, engagement) broken down by paid and organic origin. 5. User can apply date range filters to compare performance across specific campaign periods.
Alternate Flows	A1 – No paid data available: If a channel does not have any paid content in the selected period, the system displays the organic data only and notifies the user that no paid data exists for that period. A2 – API does not support paid/organic split: If the connected channel's API does not provide this breakdown, the system displays a notification explaining the limitation for that channel.
Post Condition	The user has reviewed a clear breakdown of paid versus organic performance and can use this to assess campaign ROI and inform budget decisions.

UC-09: Configure and Receive Alerts

Use Case Number	UC-09
Use Case Name	Configure and Receive Alerts
Overview	The system automatically monitors key performance thresholds and triggers alerts when those thresholds are crossed helping the team respond quickly to significant changes in their social media performance.
Actors	Co-Founder, Marketing Team Member (recipient); Admin (configuration)
Pre Condition	At least one social media channel is connected. Alert thresholds have been configured (or default thresholds apply). Alert delivery method has been confirmed with the client (dashboard notification, email, or message TBD).
Main Flow	1. System continuously monitors incoming data from connected channels. 2. System detects that a threshold has been crossed (e.g. engagement drops by 20%). 3. System generates an alert with details: channel, metric, current value, threshold, and timestamp. 4. Alert is delivered to the relevant users via the configured delivery method. 5. Alert appears in the dashboard notification centre. 6. User acknowledges or dismisses the alert.
Alternate Flows	A1 Threshold not met: No alert is generated; system continues monitoring normally. A2 Alert delivery fails (e.g. email bounces): System logs the failure and retries; alert remains visible in the dashboard.
Post Condition	Users are notified of the threshold breach. The alert is logged in the notification history for audit purposes.

UC-10: Filter Data by Different Criteria

Field	Detail
Use Case Number	UC-10
Use Case Name	Filter Data by Different Criteria
Overview	Allows a user to filter statistics and analytics views by audience demographic criteria such as geography, age, and company size, enabling the team to understand which audience segments are driving the most engagement.
Actors	Admin, Co-Founder, Marketing Team
Pre Condition	The user is logged in. At least one channel is connected and has demographic audience data available via the platform's API. The user is on the Statistics or Analytics page.
Main Flow	1. User navigates to the Statistics or Analytics page. 2. User opens the filter panel. 3. User selects one or more demographic filters (e.g. Geography: United Kingdom; Age: 25–34; Company Size: 50–200 employees). 4. System applies the selected filters and re-renders the data visualisations to reflect only the filtered audience segment. 5. User can combine multiple filter criteria simultaneously. 6. User can clear all filters to return to the unfiltered view.
Alternate Flows	A1 – Demographic data not available for a channel: If a connected channel does not expose demographic breakdowns via its API (e.g. WhatsApp Business), the system displays the filter panel but disables demographic options for that channel and explains the limitation. A2 – Filter returns no data: If the selected filter combination yields no results, the system displays an empty state message and suggests broadening the filter criteria.
Post Condition	The user has viewed statistics and analytics data filtered to a specific audience demographic segment, enabling more targeted strategic decisions.

UC-11: Optimal Posting Suggestions

Field	Detail
Use Case Number	UC-11
Use Case Name	Optimal Posting Suggestions
Overview	The system analyses historical performance data to identify patterns and surface recommendations for optimal posting times and suggested post lengths for each connected channel. These suggestions are displayed in the Analytics page or a dedicated Recommendations section.
Actors	Admin, Marketing Team
Pre Condition	The user is logged in. Sufficient historical posting and engagement data has been collected across at least one connected channel to generate statistically meaningful suggestions.
Main Flow	1. User navigates to the Analytics page or a 'Recommendations' section. 2. System analyses historical data to identify posting times correlated with highest engagement. 3. System identifies post length ranges associated with above-average performance. 4. System surfaces recommendations per channel, e.g. "Posts published on Tuesday between 18:00–20:00 receive 40% higher engagement on Instagram." 5. User can view recommendations broken down by channel and post type. 6. User can mark recommendations as actioned or dismissed.
Alternate Flows	A1 – Insufficient historical data: If fewer than a minimum number of posts have been recorded for a channel, the system displays a message indicating recommendations will become available once more data has been collected, and shows an estimated timeframe.
Post Condition	The user has reviewed data-driven posting suggestions and can apply these to their content calendar to improve future performance.

UC-12: Competitor Tracking

Field	Detail
Use Case Number	UC-12
Use Case Name	Competitor Tracking
Overview	Allows a user to add competitor social media channels to a tracked list. The system retrieves publicly available information about those channels. Due to API restrictions (particularly META), detailed metrics such as clicks and views on competitor posts are not accessible; only publicly visible signals are available.
Actors	Admin, Marketing Team
Pre Condition	The user is logged in with a role that permits competitor tracking. The competitor's social media profile is publicly accessible.
Main Flow	1. User navigates to the 'Competitor Tracking' section. 2. User enters the competitor's public social media handle or page URL. 3. System validates that the profile is publicly accessible. 4. System adds the competitor to the tracked list and begins retrieving publicly available data (e.g. follower count, posting frequency, public post content). 5. Competitor appears in the tracked list with available data populated. 6. User can view and manage the full list of tracked competitors.
Alternate Flows	A1 – Private or inaccessible profile: If the competitor's profile is private or the platform API blocks access, the system notifies the user that the profile cannot be tracked and explains why. A2 – Competitor already tracked: If the user attempts to add a competitor already on the tracked list, the system alerts them and prevents duplication.
Post Condition	The competitor is added to the tracked list and publicly available data is being retrieved and displayed for monitoring purposes.

UC-13: Competitor Trend Analysis

Field	Detail
Use Case Number	UC-13
Use Case Name	Competitor Trend Analysis
Overview	Rather than surfacing real-time competitor metrics (which are largely inaccessible via social media APIs), the system analyses patterns in competitor content over time — such as the types of posts being published most frequently — to help inform ClubArtizen's own content strategy.
Actors	Admin, Co-Founder, Marketing Team
Pre Condition	The user is logged in. At least one competitor has been added to the tracked list (UC-12) and sufficient historical data has been collected to identify trends.
Main Flow	1. User navigates to the 'Competitor Tracking' section and selects a tracked competitor. 2. System displays trend analysis for that competitor, including content type frequency (e.g. proportion of Reels vs. static posts vs. Stories over time), posting cadence trends, and any notable shifts in content strategy. 3. User can compare trends across multiple tracked competitors simultaneously. 4. User can apply a date range filter to focus on a specific period. 5. System highlights significant trend changes with visual markers or summary callouts.
Alternate Flows	A1 – Insufficient data for trend analysis: If a competitor has been tracked for too short a period to identify meaningful trends, the system displays available data and indicates when a fuller trend analysis will be possible.
Post Condition	The user has reviewed competitor content trends and can use these insights to benchmark and refine ClubArtizen's own content strategy.

UC-14: User Login and Role Management

Field	Detail
Use Case Number	UC-14
Use Case Name	User Login and Role Management
Overview	Allows authenticated users to log into the dashboard securely. Admins can create user accounts and assign roles (Co-Founder, Marketing Team) to control data access. Role permissions enforce the RBAC policy defined for the system.
Actors	Admin (role management), All users (login)
Pre Condition	The user has been issued login credentials. For role assignment, the actor must be logged in with Admin-level access.
Main Flow	1. User navigates to the login page. 2. User enters their email address and password. 3. System authenticates the credentials against the user store. 4. System identifies the user's assigned role and loads the appropriate permission set. 5. User is redirected to their role-appropriate dashboard view. 6. (Admin only) Admin navigates to the 'User Management' section. 7. Admin creates a new user account or selects an existing user. 8. Admin assigns or updates the user's role (Co-Founder, Marketing Team). 9. System saves the role assignment and applies the corresponding access permissions immediately.

Alternate Flows

A1 – Incorrect credentials: If the user enters an incorrect email or password, the system displays a generic error message (without specifying which field is wrong) and allows the user to retry or request a password reset. A2 – Account locked: After a defined number of failed login attempts, the system temporarily locks the account and notifies the user to contact their admin. A3 – Unauthorised access attempt: If a user attempts to navigate to a page outside their role permissions, the system redirects them to their permitted view and displays an access denied message.

Post Condition

The user is logged in and presented with a dashboard view appropriate to their assigned role. Admin-assigned role changes are applied immediately and reflected on next login or session refresh.